MULTIPHASE CFD SIMULATION - OIL-AIR FLOW WITH STAR-CCM+

Project: Direct oil-cooling system

Application: NOVARES e-motors

Software: STAR-CCM+

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OUTLINE

I. Introduction and context

II. Initial simulation

Geometry and mesh

Physical model and phases

Interactions and boundary conditions

Results

III. Optimised simulation

Reducing computational cost

Improving the flow

Adding heat transfer

IV. Conclusion

CONTEXT

Novares objectives

Develop cooling systems for e-motors

Focus on direct oil cooling

Simulation challenges

Complex multiphase flow (oil-air)

Multiple flow regimes

Accurate interface capture

GEOMETRY AND MESH

3D CAD geometry

Cylindrical domain

Inlet: velocity inlet

Outlet: pressure outlet

Meshing strategy

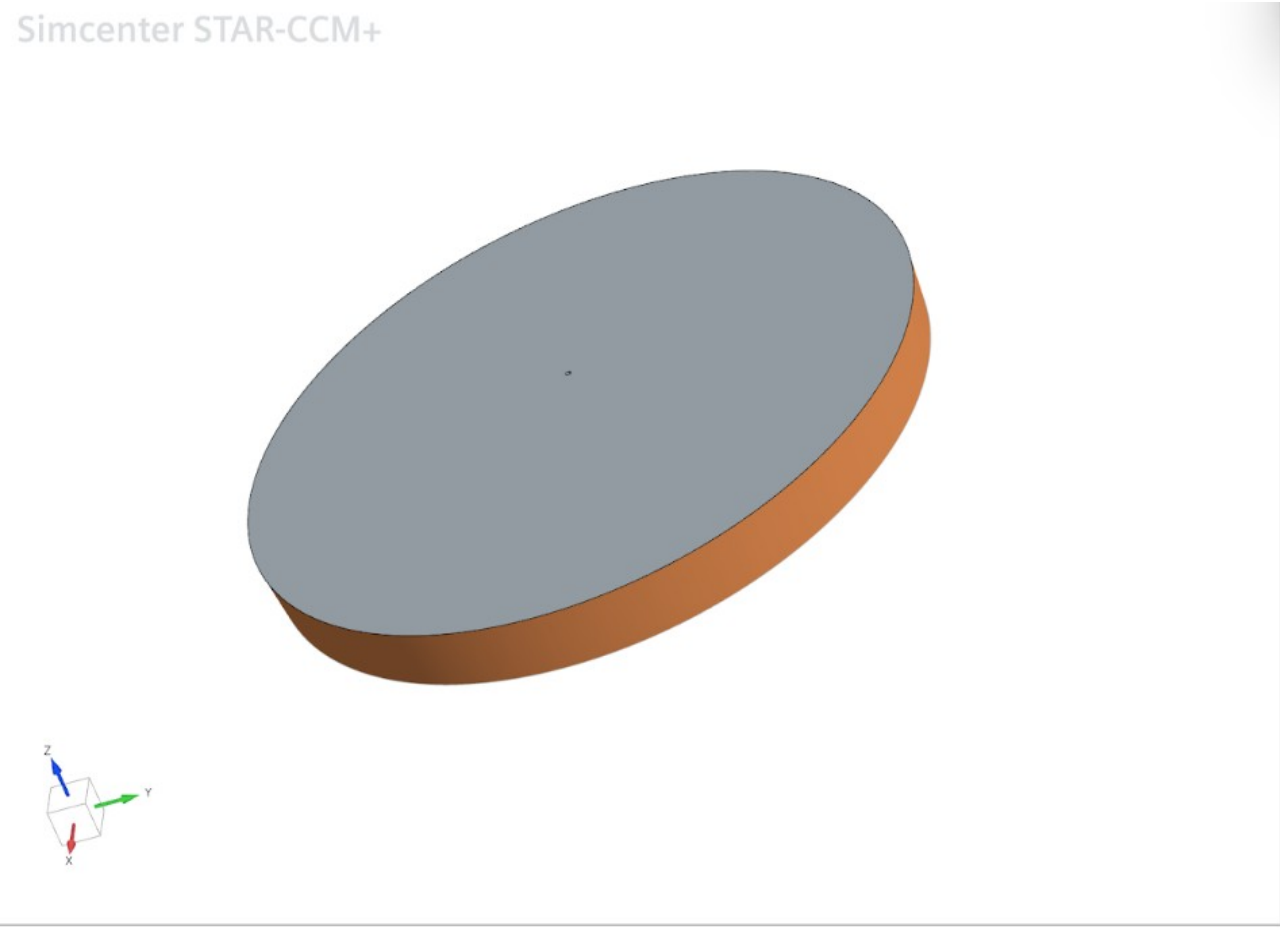
Surface Remesher + Polyhedral Mesher

Base size: 0.02 m

Local refinements:

→ Cylinder wall: 0.004 m

→ Target surface size: 0.005 m


Simcenter STAR-CCM+
Z
Y
X

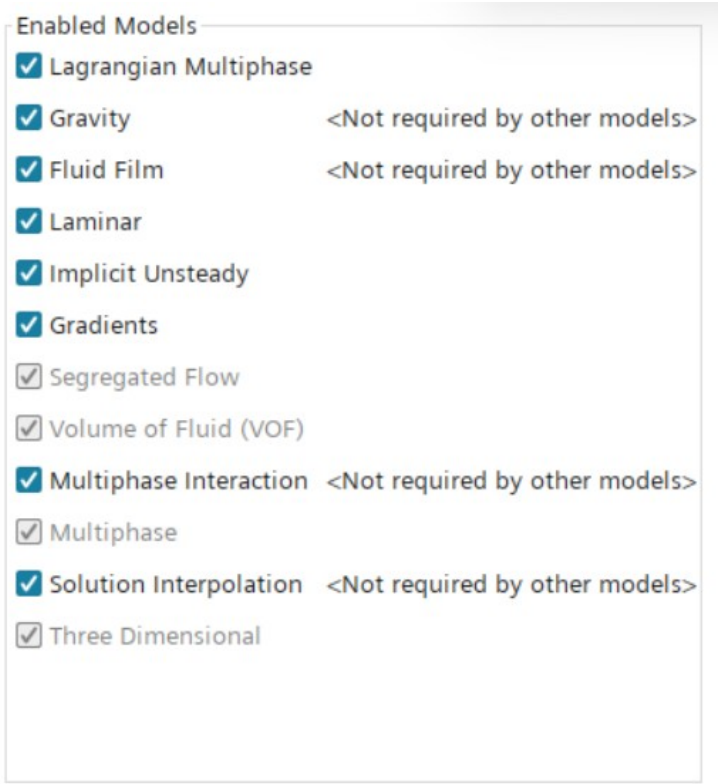

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PHYSICAL MODELS - VOF METHOD

Volume of Fluid (VOF):
- HRIC sharpening factor: 0.2
- Sharp reconstruction enabled

No adaptive mesh/time step

Phase	Type	Material	Model
Air	Eulerian	Gas	Constant Density
Oil	Eulerian	Liquid	Constant Density
Droplets	Lagrangian	Liquid	Material Particles
Film	Fluid Film	Liquid	Laminar



PHASE INTERACTIONS

VOF-VOF: Air ↔ Oil

Surface tension

VOF-Lagrangian: Oil ↔ Droplets

Resolved transition (blob criteria)

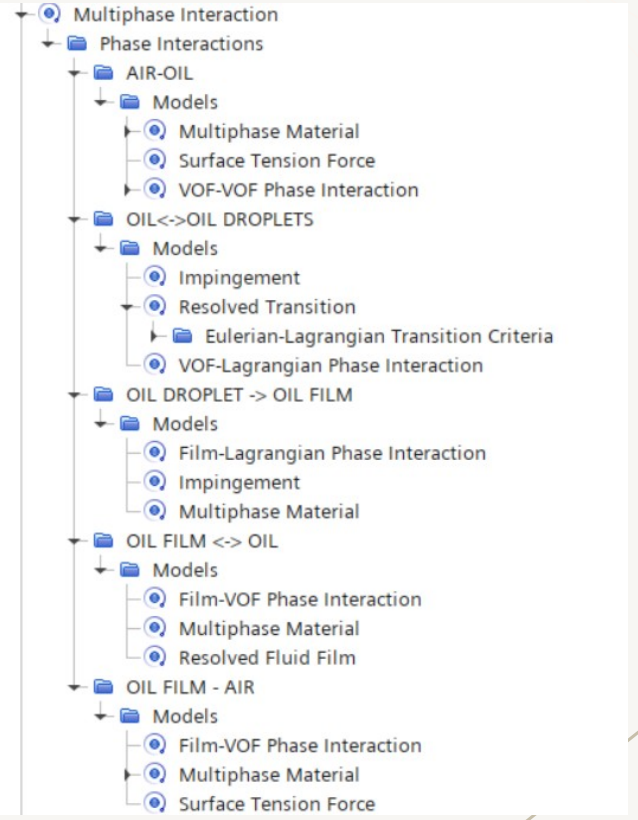
- Diameter: 0.0 - 0.05 m
- Min. sphericity: 0.7

Film-Lagrangian

Droplets impinging on the target feed the oil film

Film-VOF

Exchange between the wall film and the resolved oil phase



BOUNDARY CONDITIONS AND RUN SETTINGS

Boundary conditions

- Inlet: V = 2.5 m/s, oil volume fraction = 1.0
- Outlet: atmospheric pressure
- Shell region: oil film on Target_Surface

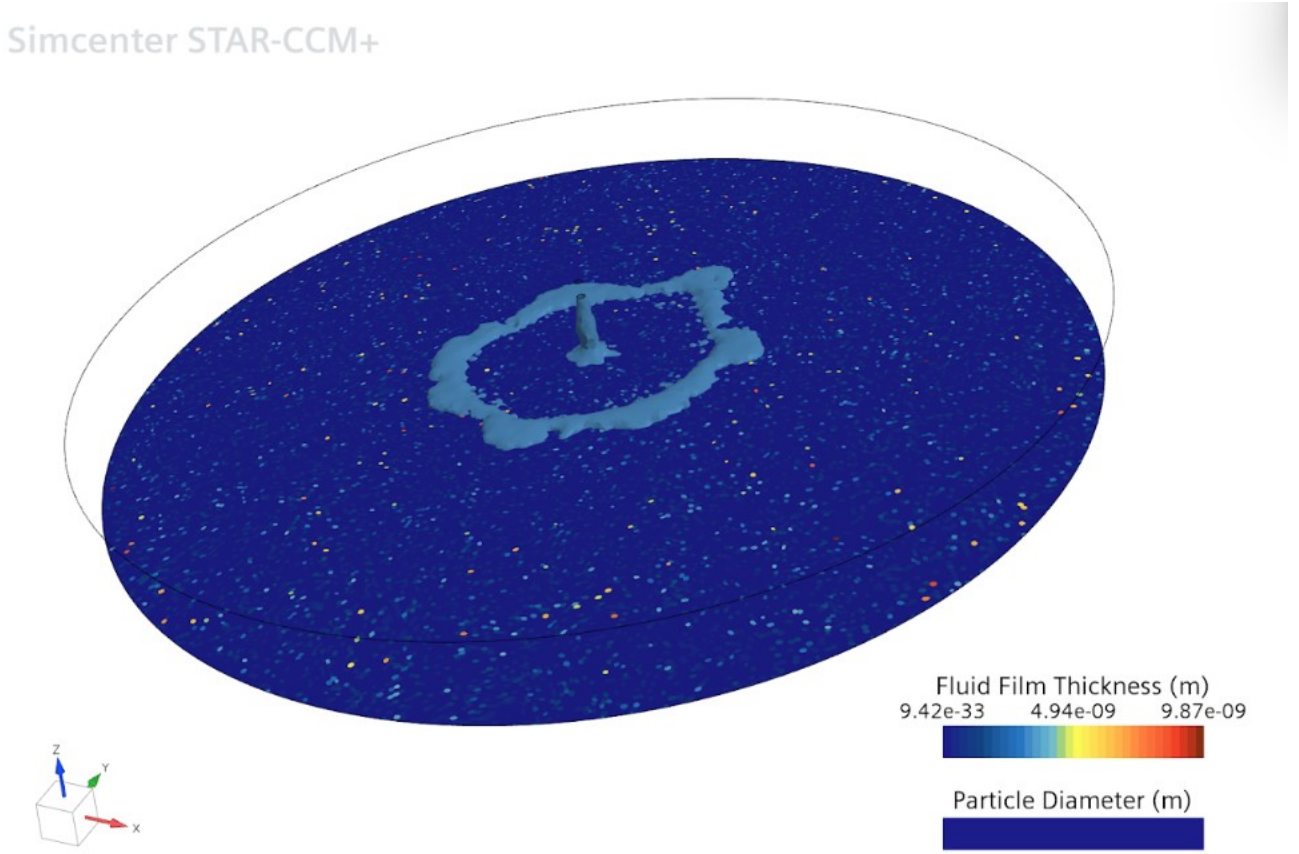
Simulation parameters

- Time step: 5.0×10^{-4} s
- Max physical time: 1.5 s

RESULTS - MULTI-REGIME VISUALISATION

Run time: 100 h
Physical time: 0.7 s
Time needed for the full simulation: ~214 h

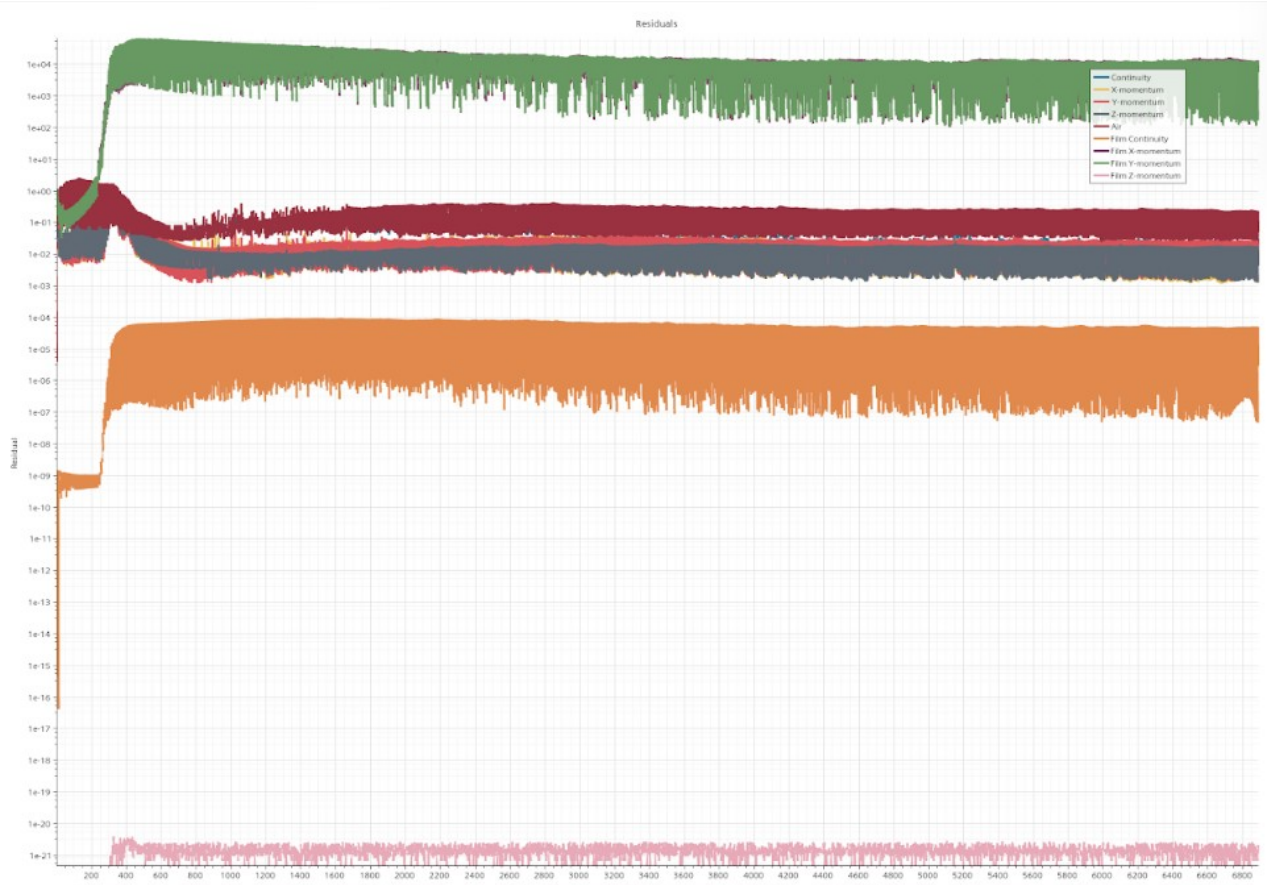
- Thick central ring of oil film
- Uneven radial deposition → uneven cooling
- Mesh adequate, but refinement at the impact point still to be added
- Adjust the sharpening factor to stabilise the interface



RESULTS - MULTI-REGIME VISUALISATION

Residuals between 10^{4} and 10^{-6} are acceptable for this kind of complex multiphase simulation

Continuity fluctuations reflect topological changes (jet break-up)



MISSION 1: REDUCING COMPUTATIONAL COST

Optimisation strategies

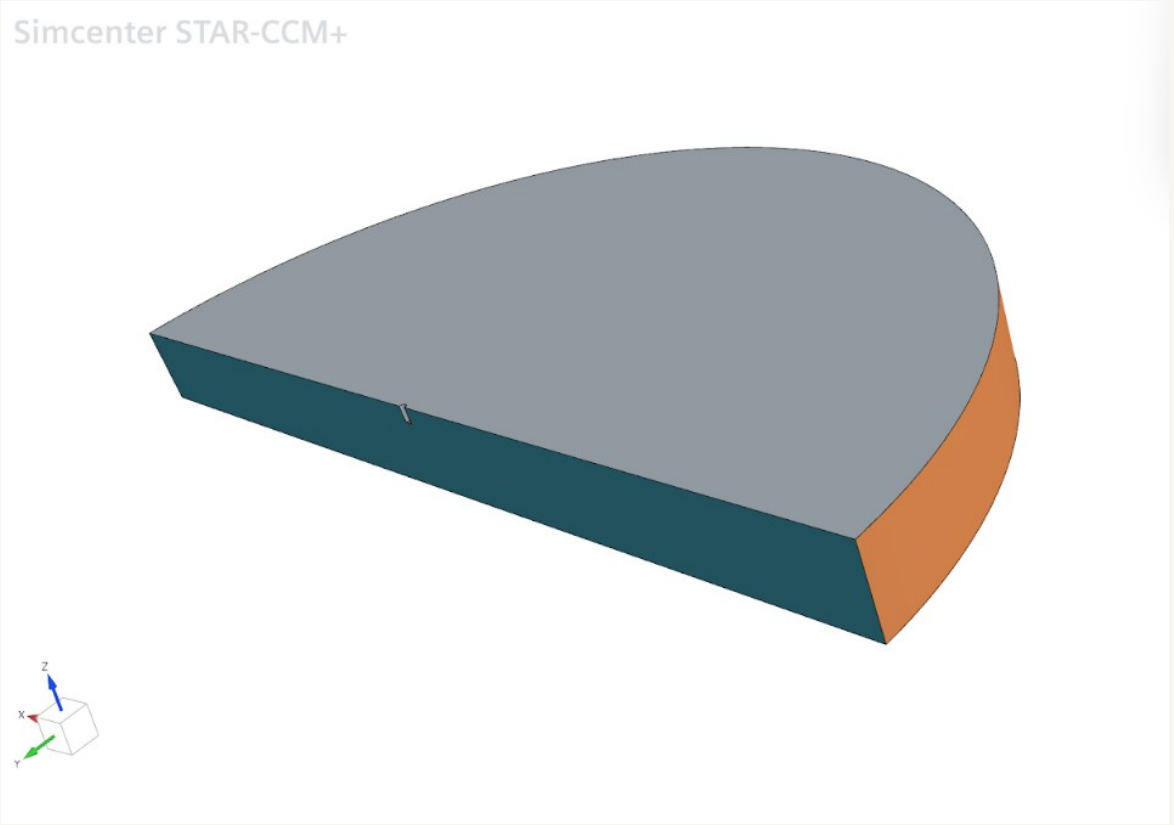
- Adaptive time step: CFL from 0.4 → 1 (largest possible time steps)
- Geometric symmetry
- Coarser initial mesh in intermediate zones
- Targeted refinement of critical zones only

Results

Run time: 50 min

Physical time: 0.4 s

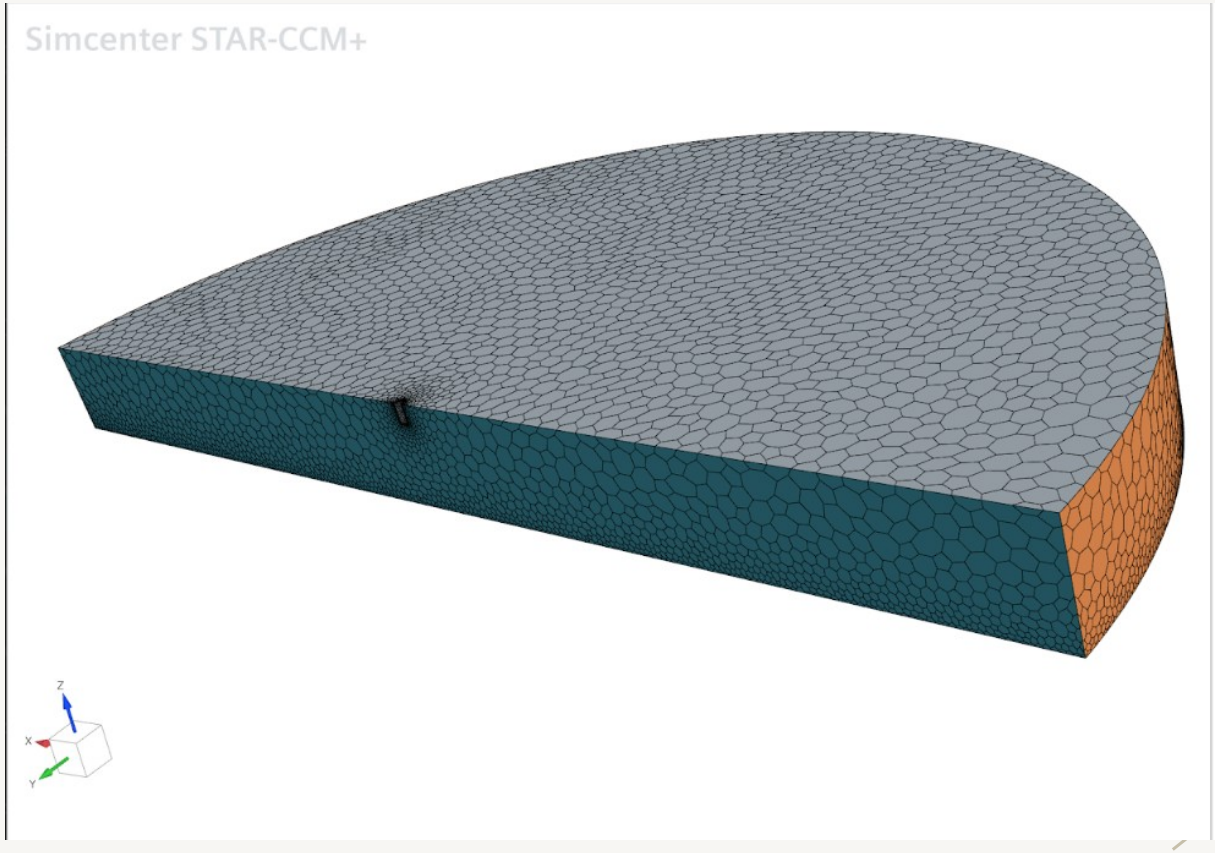
Time needed: 3.25 h



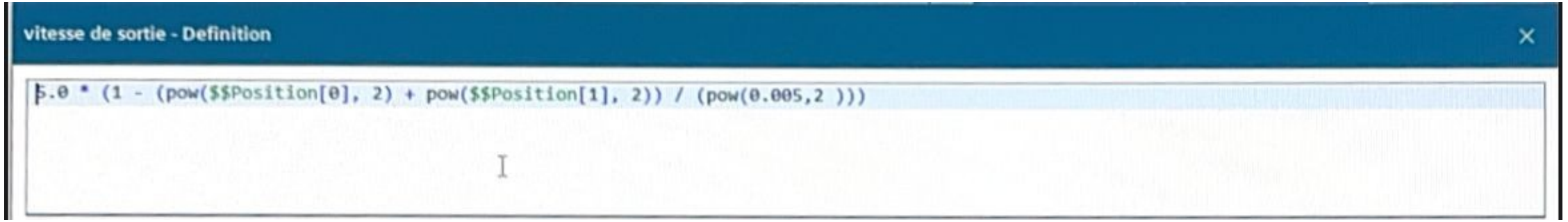
MISSION 2: IMPROVING THE FLOW

Optimisation strategies

- HRIC sharpening factor raised from 0.2 to 0.3
- Euler-Lagrange transition criteria tuned, minimum blob sphericity from 0.7 to 0.8
- Convergent nozzle geometry to improve jet cohesion and reduce pressure losses
- Field function for the inlet velocity



MISSION 2: IMPROVING THE FLOW



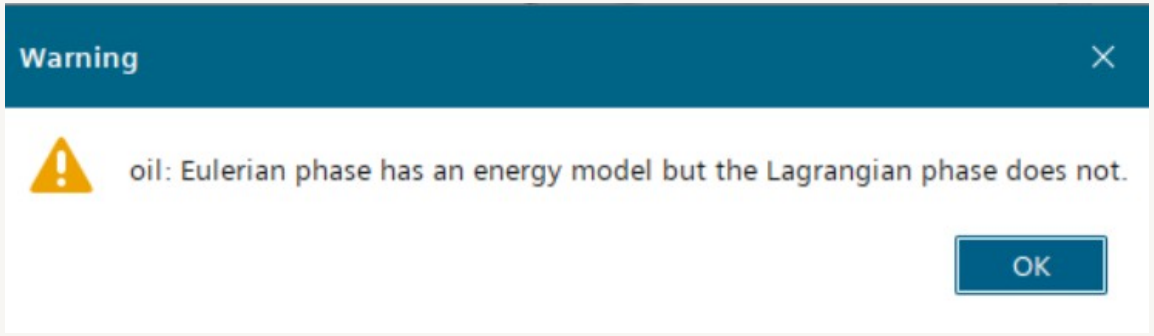
MISSION 3: HEAT TRANSFER

Thermal set-up:

- Energy model enabled (all phases)
- Thermal properties defined:
- → Thermal conductivity
- → Specific heat
- Thermal boundary conditions
- Two-way coupling (droplets)

Heat transfer coefficient computed, then post-processed in STAR-CCM+:

h = q / (T_wall - T_ref)



Not enough time to set up the "Lagrangian Energy" model

CONCLUSION

Achievements

✓ Complex multiphase VOF simulation completed

✓ Two major improvements implemented:

- Lower computational cost
- Optimised flow

Outlook

- Finish the thermal model for analysis
- Optimise the geometry

QUESTIONS?


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NOVARES
Beyond plastics